

Vaibhav Raj

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EDUCATION

Greater Noida Institute of Technology
B.Tech in Information Technology

Gr. Noida, IN
2021 – 2025

PROFILE SUMMARY

Enthusiastic and detail-driven final-year B.Tech student with a strong foundation in Artificial Intelligence and Machine Learning. Proficient in Python and widely-used ML libraries such as NumPy, Pandas, Scikit-Learn, and PyTorch. Passionate about building AI-powered solutions, with hands-on experience in computer vision, deep learning, and end-to-end machine learning pipelines. Experienced in developing intelligent systems and RESTful APIs, with a keen interest in applying advanced techniques to domains like healthcare and medical imaging. Adept at implementing research-driven models and collaborating in multidisciplinary teams to solve real-world problems.

SKILLS

Programming Languages: Python, Java

Data Science & ML: NumPy, Pandas, Scikit-learn, Matplotlib

Deep Learning & Computer Vision: TensorFlow (basics), OpenCV, CNNs

Tools & Platforms: Jupyter Notebook, Google Colab, Git, GitHub, RESTful APIs

Database Management: MySQL, MongoDB

Other Skills: Model Evaluation, Feature Engineering, Data Cleaning, Agile Development

PROJECT WORK

Smart Healthcare: Liver Segmentation Image Analysis Using AI | [GitHub Link](#)

- Developed a deep learning-based liver segmentation system using **MONAI** and **PyTorch**, leveraging U-Net architecture for medical image analysis.
- Preprocessed and trained on CT/MRI scan datasets, applying MONAI transforms for high-precision segmentation tasks.

AQI Prediction: Multivariate Time Series Forecasting using LSTM | [GitHub Link](#)

- Designed and implemented an LSTM-based deep learning model to predict Air Quality Index (AQI) using multivariate time series data.
- Utilized multiple environmental parameters (e.g., PM2.5, temperature, humidity) as features, applying data normalization and sequence generation for model training.
- Built and trained the model in Python using TensorFlow/Keras, achieving accurate short-term AQI forecasting to support environmental monitoring.

CERTIFICATIONS

- Java Certification from MSME (Micro, Small & Medium Enterprises, Government of India)
- AI for Beginners from HP foundation.
- SQL certificate from HackerRank.